

OSSP

Practical- Assignment1

Name: Praneeth Kumar Gadila

Roll num: 2520030257

1.

```
GNU nano 8.7.1                                practical1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[20];

    printf("Enter command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

```
praneethgadila@praneethgadila:~$ nano practical1.c
praneethgadila@praneethgadila:~$ nano practical1.c
praneethgadila@praneethgadila:~$ gcc practical1.c
praneethgadila@praneethgadila:~$ ./a.out
Enter command: ls
Parent PID: 775
Child PID: 776
a.out  dir2  docs  exec3.c  f8      family_forck.c  forkk.c  kiran.c  openn.c  pipee.c  pran2  public.txt
backup dir5  exec1.c  f1      f5      file1          gb.c     linux_task4  permission_lab  practical1.c  pran3  script.sh
config dir6  exec2.c  f2      f8      file1          grepp.c  logs        permission_lab  practical1.c  private.txt  src
dd.c   dir65 exec3.   f3      family.c  fipec          haadifile nano.1223.save pipee  practical1.c  prog.c  student_portal
Child completed
praneethgadila@praneethgadila:~$ |
```

2.

```
praneethgadila@praneethgadila:~$ uname
Linux
praneethgadila@praneethgadila:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13450HX
CPU family: 6
Model: 183
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
Stepping: 1
BogoMIPS: 5222.40
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bti1 avx2 smep bmi2 erms invpcid rdseed adx saop clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsave avx_vnni vnni unip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 384 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 10 MiB (8 instances)
L3: 20 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
```

```
NUMA node0 CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected

praneethgadila@praneethgadila:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro

praneethgadila@praneethgadila:~$ ps
PID TTY TIME CMD
333 pts/0 00:00:00 bash
510 pts/0 00:00:00 ps
```

```
praneethgadi@praneethgadi:~$ top
top - 05:27:36 up 5 min, 1 user, load average: 0.11, 0.05, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7783.7 total, 6629.2 free, 598.3 used, 788.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7185.3 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM     TIME+ COMMAND
 522 praneet+ 20   0   8172  6136  4820 R   0.3   0.1   0:00.07 top
    1 root      20   0  24984 15668 11564 S   0.0   0.2   0:00.75 systemd
    2 root      20   0   3180  2204  2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    7 root      20   0   3180  2044  1940 S   0.0   0.0   0:00.00 init
   47 root     19  -1  50404 16524 15240 S   0.0   0.2   0:00.13 systemd-journal
   79 systemd+ 20   0  22416 14508 12008 S   0.0   0.2   0:00.06 systemd-resolve
   87 root      20   0  35424 12416  9268 S   0.0   0.2   0:00.29 systemd-udev
  108 root      20   0   2888  1952  1820 S   0.0   0.0   0:00.05 chronyd-starter
  109 root      20   0   4472  3144  2896 S   0.0   0.0   0:00.00 cron
  110 message+ 20   0   8812  5360  4688 S   0.0   0.1   0:00.04 dbus-daemon
  162 root      20   0  48096 29916 14780 S   0.0   0.4   0:00.26 networkd-dispat
  167 root      20   0  18436  9396  8148 S   0.0   0.1   0:00.12 systemd-logind
  176 root      20   0 1792300 15588 12740 S   0.0   0.2   0:00.12 wsl-pro-service
  186 syslog    20   0  220548 5924  4632 S   0.0   0.1   0:00.07 rsyslogd
  267 _chrony     20   0  23736 11276  7240 S   0.0   0.1   0:00.06 chrony
  268 _chrony     20   0  12096  2424  1780 S   0.0   0.0   0:00.00 chrony
  280 root      20   0 123460 32788 18176 S   0.0   0.4   0:00.18 unattended-upgr
  283 root      20   0   5492  2728  2544 S   0.0   0.0   0:00.00 agetty
  331 root      20   0   3196  1112   980 S   0.0   0.0   0:00.00 SessionLeader
  332 root      20   0   3196  1248  1108 S   0.0   0.0   0:00.01 Relay(333)
  333 praneet+ 20   0   6268  5524  3856 S   0.0   0.1   0:00.00 bash
  334 root      20   0   8648  5352  4708 S   0.0   0.1   0:00.01 login
  378 praneet+ 20   0  22756 13360 10980 S   0.0   0.2   0:00.12 systemd
  380 praneet+ 20   0  22924  4896  2888 S   0.0   0.1   0:00.00 (sd-pam)
  406 praneet+ 20   0   6136  5408  3848 S   0.0   0.1   0:00.03 bash
```

- **Linux** OS acts as a link between computer hardware and applications.
- Linux manages the CPU using process scheduling.
- Linux manages storage using files, directories, and file systems.
- **uname** is used to check the Linux kernel and system information.
- **lscpu** gives details about the CPU, cores, threads, and architecture.
- **lsblk** shows the storage devices, disks, SSDs, and partitions.
- **ps** is used to see the running processes and their PIDs.
- **top** shows the real-time CPU, memory, and process usage.